

STRING

System for Traceable Reasoning Integrity
through Networked Graphs

팀명: CPR (Track 1: CR)

팀원: 김동현, 노승현, 조나현, 최현민

GitHub:



Rationales

➤ Background

- Formal solvers verify logical validity **after** reasoning has been translated into formal expressions. Therefore, in real-world LLM outputs, a major bottleneck is the **pre-verification stage**.

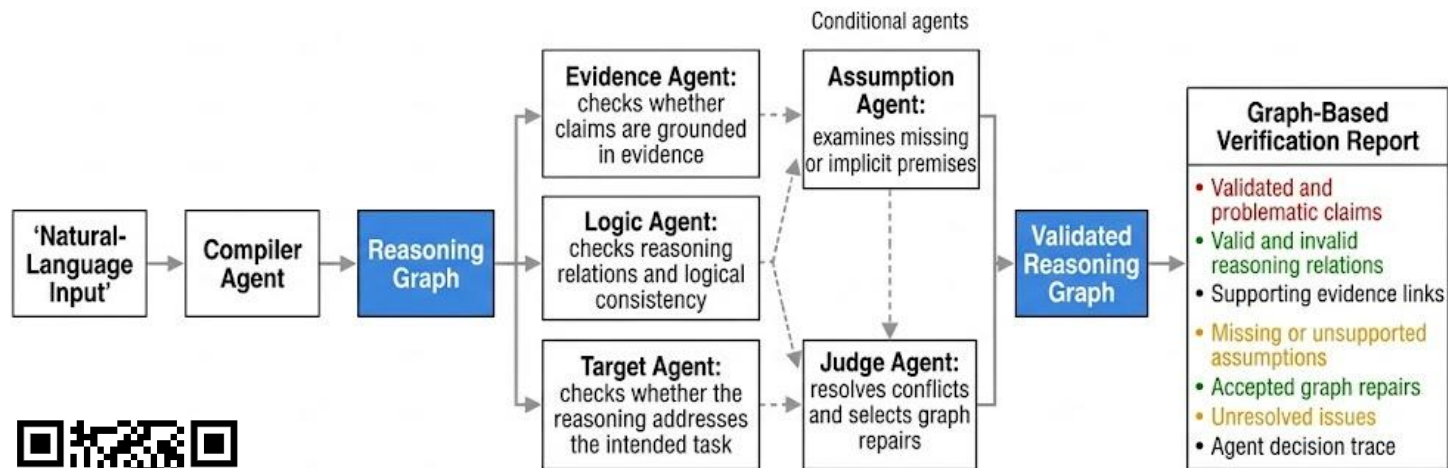
➤ Related Work

- VeriCoT: Chain of Thought steps & symbolic reasoning
 - Graph of Thoughts: represents as a graph
 - Multi-Agent Debate & Autogen: flexible interaction patterns
 - CRITIC: independent feedback and iterative correction in LLM verification
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- Positioning of STRING: combines a graph-based intermediate representation with role-specialized verification, focusing on semantic grounding, reasoning integrity, task alignment, and implicit assumptions

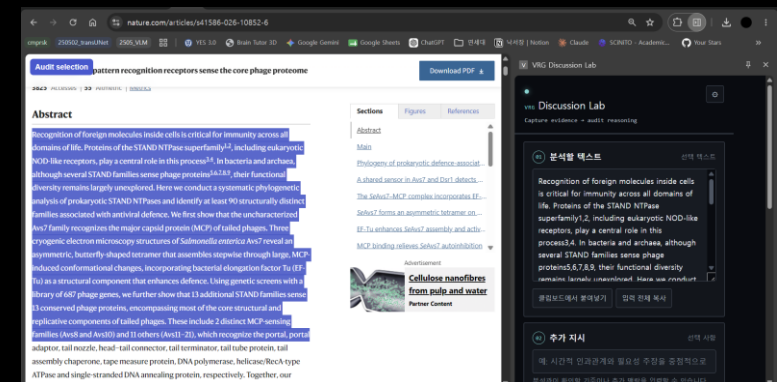
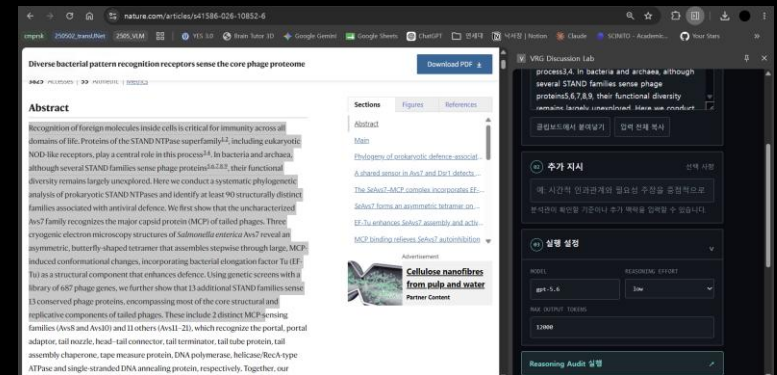
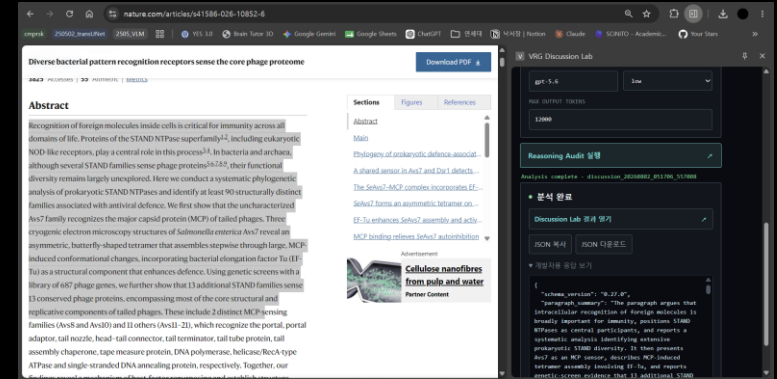
Model Overview

- STRING Architecture
- Chrome Extension

STRING — System for Traceable Reasoning Integrity through Networked Graphs



Specialized agents inspect different error dimensions; they are not identical voters.



Application

- AI Hallucination Detection
- RAGTruth Dataset

CLAIMGRAPH / CROSS-DOCUMENT VALIDATION

Source ↔ Response graph

출처에 없는 표현, 한정적으로 지지되는 표현, 직접 근거가 있는 주장을 node 단위로 분리합니다.

case 13931 · direct_case_id_review
introduction to visual merchandising and relation to the consumer

Verdict summary

응답 전체가 틀린 것이 아니라, R1의 unsupported 표현과 R7의 qualified 표현을 근거 있는 주장과 분리해 확인합니다.

RAW F10.0000BALANCED F11.0000

Graph stage

6-Agent validated

Demo focus · unsupported / qualified

Clear / Esc

Six-Agent validated

SOURCE EVIDENCE

source passage

[e1] introduction to visual merchandising and relation to the consumer

[e2] passage 1:Introduction to Visual Merchandising. The visual merchandiser is aware of the retail strategy and the company goals of the retailer in order to project the same in the store. Earlier retailers used advertising as the only media to attract customers to the store.

[e3] passage 2:ROLE OF MERCHANDISER. Marketing brings horse to the water but visual merchandising makes the horse drink G. Surrender

RESPONSE

candidate span fallback enabled

Based on the provided passages, here is my answer. The passage highlights the key role of visual merchandisers in creating an immersive and engaging retail experience for customers. They are responsible for bringing the retail strategy and company goals to life in the store through effective display and presentation of products. The passage also emphasizes the importance of visual merchandising in influencing consumer behavior, particularly in the context of shop windows. The second passage discusses the role of merchandisers in making the product interesting and appealing to consumers, drawing a comparison between the role of marketers (who bring

CLAIMGRAPH · SOURCE / RESPONSE / CROSS

8 source nodes · 7 response nodes · 8 cross edges

CLAIMGRAPH / RESPONSE WEAKNESS REVIEW

Source ↔ Response graph

응답이 먼저 제시된 근거와 뒤에서 스스로 부정한 문장을 node 단위로 비교합니다.

case 11908 · direct_case_id_review
how do automotive technicians get paid

Verdict summary

R3의 Mississippi 최저임금 수치는 출처에 없고, R6-R7-R8은 앞선 response와 source evidence를 부정하거나 과도하게 일반화합니다.

RAW F10.3500BALANCED F11.0000

Graph stage

6-Agent validated

Demo focus · response weaknesses

Clear / Esc

Six-Agent validated

SOURCE EVIDENCE

evidence units

[e1] how do automotive technicians get paid

[e2] passage 1:In short, whether mechanics and technicians are entitled to overtime wages depends on (a) where they work (auto dealer or repair shop), (b) how they are paid (commission or not), (c) how much they make (regular hourly rate and commissions), and (d) the state in which they work.d 1554, finding that certain automotive mechanics and technicians paid on a flat-rate or flag-rate are entitled to overtime under the FLSA.

RESPONSE

claim spans and conclusions

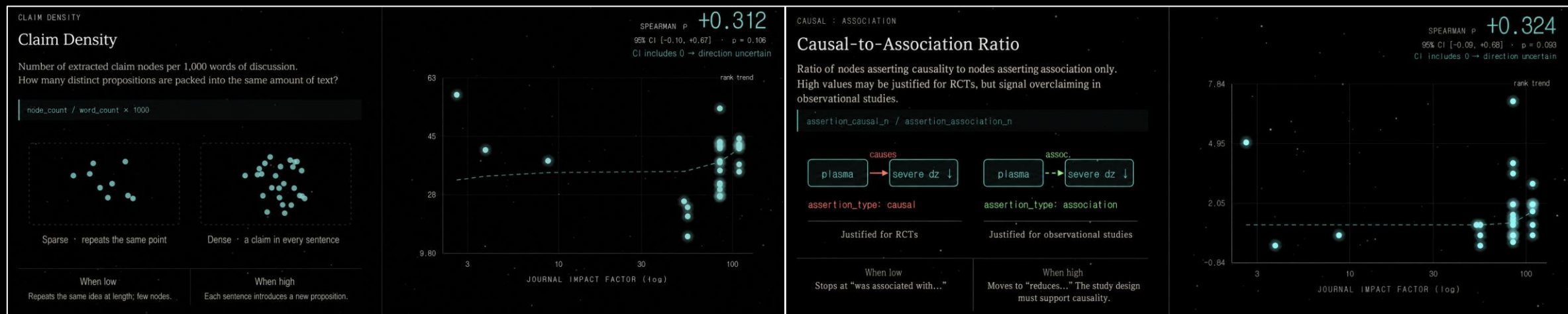
Based on the provided passages, automotive technicians can get paid in different ways, including hourly and commission-based pay. The specific amount of pay varies by location, with the highest average pay in Alaska (\$23.70 per hour or \$49,400 per year) and the lowest average pay in Mississippi (\$18.60 per hour or \$38,900 per year). Additionally, some automotive technicians may earn more working in related industries such as aerospace products and parts manufacturing, which pays an average of \$32 per hour or \$66,300 per year. However, the passages do not provide specific information on how automotive technicians get paid or their typical salaries.

CLAIMGRAPH · SOURCE / RESPONSE / CROSS

12 source nodes · 8 response nodes · 16 cross edges

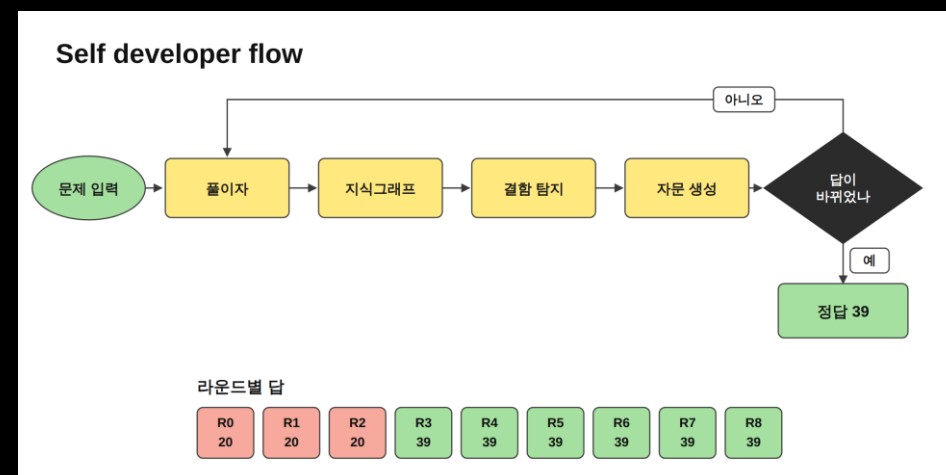
Application

➤ Correlation with Impact Factor



➤ Math Problem Solver

- Architecture: + Self-Revision loop
- Dataset: AIME problems that GPT-5.4 nano could hardly solve



Limitations

➤ Symbolic Reasoning Integration

- The Math Problem Solver currently relies primarily on LLM-based reasoning.
- Integrating a symbolic constraint solver such as Z3 could improve arithmetic consistency, constraint verification, and final-answer reliability.
- A hybrid **LLM–Z3 architecture** would allow the LLM to formulate the reasoning process while Z3 verifies formally expressible conditions.

➤ Optimization Strategies

- Our current **optimization strategy** mainly relies on:
 - Activating agents only when needed
 - Using smaller models such as GPT-5.4 nano and GPT-5.4 mini
- Additional approaches could include: Confidence-based routing, Early stopping, Prompt and context compression, Parallel agent execution, Model distillation

References

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